

Blueprint: The Case I/II engine and Washington
9.5 descent (flt-vandiver)

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Abstract

This blueprint maps `flt-vandiver`, the engine of the development: it reduces `FermatLastTheoremFor` p to two finite certificates, discharging Case II by a machine-checked instance of Washington’s Theorem 9.5 small-witness descent and Case I by a Bernoulli or Sophie Germain criterion. Regularity is never used; the weaker input $p \nmid h^+$ is supplied by a Q_i cyclotomic-unit certificate.

Two inputs are consumed from sibling libraries and appear here as external leaves: the cyclotomic-unit index $[E : C] = h^+$ (from `flt-cyclotomic-nt`) and Hilbert’s Theorem 94 with Kummer-extension unramifiedness (from `flt-regular`).

Nodes are linked by `\lean` to their declarations (namespaces `FltVandiver`, `FltVandiver.Descent95`); this is a milestone-level map of the engine, centered on the Case II descent.

0.1 The Q_i certificate and the Vandiver bridge

Definition 0.1.1 (Q_i / Vandiver certificate). *The Boolean certificate `vandiverCert p l t` asserting that the Q_i cyclotomic-unit test fires at every even index for the auxiliary pair (ℓ, t) .*

Definition 0.1.2 (Fast sliced certificate). *A pure-`Nat` reformulation `vandiverCertFast` of the certificate, split into slices, that `native_decide` can evaluate even at $p = 2124679$.*

Definition 0.1.3 (Unit index $[E : C] = h^+$ (external)). *Washington’s Theorem 8.2, `cyclotomic_unit_index` in `flt-cyclotomic-nt`: the index of the cyclotomic units equals h^+ . It turns the Q_i test into a statement about $p \nmid h^+$.*

Theorem 0.1.4 (Q_i certificate $\Rightarrow$ Vandiver). *If `vandiverCert p l t` holds then $p \nmid h^+$: the Q_i test firing at every even index rules out each eigen cyclotomic unit being a p -th power.*

Definition 0.1.5 (Small-witness certificate). *`SmallWitnessCert p` packages the pair (ℓ, t) with $\ell < p^2 - p$ and the Q_i certificate — the single finite input that powers both the Vandiver conclusion and the Case II descent.*

0.2 Case II: the Washington 9.5 small-witness descent

Definition 0.2.1 (Hilbert 94 / Kummer unramifiedness (external)). *Hilbert’s Theorem 94 and the unramifiedness of the relevant Kummer extension, consumed from `flt-regular`. These seed the $\ell \mid \xi$ invariant that the descent tracks.*

Theorem 0.2.2 (Kummer extension unramified). *The Kummer extension attached to a putative Case II solution is unramified away from ℓ , the input the descent invokes a second time to seed its invariant.*

Theorem 0.2.3 (Situation from a rational solution). *A putative Case II solution, normalized to the real cyclotomic field, produces a “situation” of Washington 9.5 equipped with a measure $m \geq p(p-1)/2$.*

Theorem 0.2.4 (Principality from Vandiver (Washington 9.6)). *Under $p \nmid h^+$ alone, the ideal factors of the situation are principal — the step that classically needed full regularity, here powered by the certificate.*

Theorem 0.2.5 (One descent step). *From a situation the descent produces a new situation at a smaller parameter, via the mod- ℓ cancellations of Washington’s Lemmas 9.6–9.7.*

Theorem 0.2.6 (The step decreases the measure). *The descended situation has strictly smaller measure — the well-foundedness that makes the descent terminate.*

Theorem 0.2.7 (No minimal situation). *No situation of minimal measure can exist; hence Case II admits no solution.*

0.3 Case I

Definition 0.3.1 (Case I hypothesis). *CaseI Holds* p : no Case I solution ($p \nmid xyz$) exists. It is a hypothesis of the modular engine, discharged below by a certificate.

Theorem 0.3.2 (B_{p-5} route to Case I). If $p \nmid B_{p-5}$ then Case I holds — Kummer’s 1857 refinement, used at the Wolstenholme primes where $p \mid B_{p-3}$ silences the primary criterion.

Definition 0.3.3 (Sophie Germain certificate). *sgCert* pq : the Boolean certificate that an auxiliary prime q meets Sophie Germain’s conditions for p .

Theorem 0.3.4 (Germain route to Case I). If *sgCert* pq holds for some auxiliary prime q , then Case I holds — the certified route at the two Wolstenholme primes.

0.4 The crown theorems

Theorem 0.4.1 (Engine, modular form). If *SmallWitnessCert* p holds and Case I holds at p , then *FermatLastTheoremFor* p : Case II is closed by the descent, Case I taken as hypothesis.

Theorem 0.4.2 (Engine, discharged form). Given the Q_i certificate, the size bound $\ell < p^2 - p$, and the irregularity certificate with $p-3 \notin L$, *FermatLastTheoremFor* p holds with nothing about Case I assumed — the non-Wolstenholme headline.

Theorem 0.4.3 (Reduction to two open problems). If every prime has a small-witness certificate and every odd prime has a Sophie Germain auxiliary, then Fermat’s Last Theorem holds — with no modularity input. This is the paper’s headline reduction; both hypotheses are classically open.